

PROCESS OR MATERIALS REVIEW REQUEST FORM

Request Title: DNA/Protein Hydrogel examination

Requestor: Kristina Kovac

Date: 07/20/20

Non-SNF staff, please complete the following:

Badger ID: kkovac

Company name/PI: Ashby Morrison

Send completed form to: snf-promcommittee@lists.stanford.edu

1. **The chemical or material.** Please provide all common names, trade names, and CAS numbers where appropriate. Include an MSDS, if available; or provide the reason, if not. Make sure to include information for any new secondary chemicals (such as a developer for a new resist). Read the MSDSs as well as the [Stanford Chemical Storage Groups](#) to determine the Storage Group Identifier and Main Hazard Class of your chemical/material.

Recombinantly purified heterochromatin protein 1 alpha - HP1alpha (CBX5) that undergoes liquid-liquid phase separation upon binding to DNA will be mixed with 5000bp circular plasmid DNA. All proteins and DNA were made in the Morrison Lab and there is no MSDS for these items. DNA is dissolved in ultra-pure water. Protein is dissolved in buffer solution containing 75mM potassium chloride (CAS 7447-40-7, molar mass 74.55g/mol), 20mM HEPES (4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid, CAS 7365-45-9) which is a zwitterionic buffering agent required to maintain physiological pH, and 2mM DTT (Dithiothreitol) as redox reagent (CAS 3483-12-3). Weak hydrogel is formed by mixing 400uM (88ug) of protein with 92nM (3ug) of DNA in 10 ul volume in the protein buffer (components described above). All components are non-toxic, non-hazardous, non-flammable and non-infectious. Spills are unlikely to occur due to small volumes, however any spills can be cleaned with regular lab wipes. Components need to be mixed directly prior to measurement and incubated for 10 min at the room temperature. Mixing can be performed on lab bench and fume hood is not necessary.

2. **Vendor/manufacture info:** address and phone number, website URL.

All materials were produced by the requestor:

Kristina Kovac, Dept. of Biology, Stanford University

Gilbert Building, Room 328, 371 Serra Mall, Stanford, CA 94305

Lab phone: (650)724 0422

3. **Reason for request:** Please give serious thought to this. If you have any process information (application notes from the vendor, protocol from another lab, experimental methods section of an article), please include it, preferably as attachments. Ask yourself these questions: Is this the latest procedure? Are there newer/safer alternatives that will also work for my project? Will any of the current SNF approved chemicals and materials work for me?

Experiments have shown that when mixed with DNA, HP1alpha undergoes through liquid-liquid phase separation in which solution separates in two distinct phases, one that is enriched with DNA/protein and other that is depletes. We hypothesize that phase separation might provide protection from DNA damaging UV light. We have already measured transmittance and now we are interested in measuring absorption and/or reflection, for which we want to use JASCO UV-spectrophotometer.

4. **Process Flow:** Please provide a detailed process flow description on how and where you proposed to use this chemical. This should include **all Lab equipment** to be used for processing your wafers once your new chemical or material has been used (even if your new material is a film that is removed, it may still pose potential contamination concerns.) Make sure to include wet benches. Please note that if the chemical/material is to be used in any the "clean" equipment, purity specifications will be needed. This is most important for chemicals/material that are not normally used for VLSI device fabrication. To be allowed into a "clean" tool, the material should MOS grade or better.

We are planning to use JASCO UV-spectrophotometer. Samples will be brought on ice and mixed directly prior to measurement. Samples need to be mixed on wet lab bench. Mixing of samples will be performed in microcentrifuge tube. Micropipettes and tips will be used for mixing and will be brought in by the requestor. Any spills can be removed with regular lab wipes. We will be measuring 5 liquid hydrogel samples.

5. **Amount and form.** How much will you bring in? Is it solid, powder or liquid? (Note: as a general rule, powders are not permitted in the cleanroom.) Do you need to mix it to use it?

All samples are liquid. Small volumes of buffer and DNA stock solution for mixing will be brought in (100ul). Protein stock will be brought in 50ul volume. Mixing will be performed on wet lab bench in 10 ul volume

6. **Storage:** Will you be storing your chemical/material at SNF? If so, please note any potential reactivity's (this should be on the MSDS). [Storage groups](#) A, B, D and L are stored in the yellow solvent cabinet in the furnace support area, while [storage groups](#) C, E, F and G are stored on top of one of the Pass-through Carts. Ensure your chemical container or material is properly labeled. If there is no available room, it must be stored by in the bulk storage

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area. You will then need to obtain it from receiving area personnel each time you want to use it and return it to them when you are finished using it (or each time you leave the lab). Note that there is no storage of chemicals/materials in the processing lab or at any wet bench.

N/A

7. **Disposal:** How will you dispose of any waste or excess chemical or material? In your discussions with experts and vendors, try to determine the best way to dispose of your spent chemicals and by-products. Please refer to the [SNF Lab members Safety Manual](#) for the different methods of waste disposal that are available in the lab.

N/A

To be completed by the PROM COMMITTEE:

Special handling requirements:

On-going approval or staff interaction requirements:

Approved By/Date: